

LangChain Components

1 Introduction & Recap

This explains the **core components of LangChain** and how they work together to build GenAI applications efficiently. It builds on the previous concept by focusing on architecture-level understanding.

Example: Before building an AI chatbot, you must understand models, prompts, memory, and agents — which are the fundamental building blocks.

2 LangChain Components Overview

LangChain is structured into reusable components that standardize how LLM applications are built. These components include models, prompts, chains, indexes, memory, and agents.

It allows developers to connect different modules without rewriting logic for each new project.

Example: You can reuse the same memory component across multiple chatbot projects.

3 Models

Models are the core LLM interfaces that generate responses. LangChain provides a unified way to connect to different LLM providers.

It abstracts model APIs so developers can switch models easily without changing application logic.

Example: You can switch from OpenAI GPT to another LLM model by updating configuration instead of rewriting your code.

4 Prompts

Prompts define structured instructions sent to the model. LangChain provides prompt templates to standardize and manage dynamic inputs.

Good prompt design improves accuracy and consistency in AI responses.

Example:

Prompt Template:

“Answer the question based on context: {context}. Question: {user_question}”
→ The system automatically fills in values before sending to the model.

5 Chains

Chains connect multiple components into a workflow. Instead of calling a model once, chains allow sequential processing steps.

They automate multi-step reasoning or processing pipelines.

Example:

User input → Clean text → Retrieve data → Send to model → Format response

All steps run automatically using a chain.

6 Indexes

Indexes store and organize external data so the model can retrieve relevant information efficiently. They are mainly used for Retrieval-Augmented Generation (RAG).

Indexes enable semantic search over documents.

Example:

You upload a PDF → It is converted into embeddings → Stored in a vector database → The system retrieves relevant sections when asked.

7 Memory

Memory allows the system to remember past interactions in a conversation. It stores chat history or contextual data to maintain continuity.

Without memory, LLMs treat every query independently.

Example:

User says: “My name is Ali.”

Later asks: “What is my name?”

The system recalls stored memory and responds correctly.

8 Agents

Agents are intelligent decision-makers that choose which tools or actions to use based on user input. They can call APIs, search tools, or databases dynamically.

Agents increase automation and flexibility in AI systems.

Example:

User asks: "What is today's weather in Lahore?"

Agent → Decides to call a weather API → Fetches data → Returns answer.
